

Civic.ai - A Citizen Centric AI-Powered Government Service Assistant

by

G.Araf Ahmed

22101532

Mohammad Zunaïd

24241264

Md Shihab Sarker

22101516

Fahim Foysal Abir

22101543

Tapu Mohammad Fahad

24241263

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Department of Computer Science and Engineering
Brac University

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Declaration

This manuscript is prepared for the authors' review and signature. The authors must confirm that the work submitted is their own, that external sources are appropriately acknowledged and that the final document complies with the university's academic-integrity and tool-disclosure requirements.

Author confirmation and signatures

G.Araf Ahmed (22101532)

Mohammad Zunaid (24241264)

Md Shihab Sarker (22101516)

Fahim Foysal Abir (22101543)

Tapu Mohammad Fahad (24241263)

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Supervisor	Name, designation and signature to be confirmed
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Co-supervisor, if applicable	Name, designation and signature to be confirmed
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Thesis coordinator	Name, designation and signature to be confirmed
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Head of department	Name, designation and signature to be confirmed
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Ethics Statement

This project develops an informational assistant from collected government-service documents and prepared text derivatives. It does not issue official decisions, authenticate identities, process payments or approve applications. Public availability is not treated as an automatic redistribution licence. Original documents, derived text and source links must retain appropriate provenance and usage restrictions.

No completed participant study, independently adjudicated expert benchmark or ethics-board approval is claimed in this manuscript. Informal developer and domain-informed observations are distinguished from systematic human evaluation. Personal identification numbers, scans and private application records are not needed for the reported fixed-question experiments and should not be collected by a public deployment without appropriate safeguards.

The project used AI-assisted coding, debugging and manuscript preparation. The authors remain responsible for reviewing the implementation, references, analysis and final submission. Model-generated answers and automatic judges are not authoritative interpretations of government rules. Review flags and incomplete outputs are disclosed rather than represented as verified advice.

Abstract

Accessing government services in Bangladesh is often difficult due to confusing procedures, lack of clear and centralized information and reliance on middlemen. This prevents many citizens from accessing these services efficiently, resulting in unnecessary time delays and financial burdens. This thesis aims to solve this problem by developing an AI-Powered interactive system that provides citizen centric guidance for certain government services which will act as a digital bridge between citizens and government services. This system will inform users about required procedures and credentials of the respective services by offering dialogue and question-answering sessions. The outcome is a user-friendly, accessible system that empowers citizens with correct information, reduces dependency on intermediaries and simplifies interactions with government offices.

Keywords: Artificial Intelligence (AI), Government Services, Intermediaries, AI-Powered interactive system, Citizen centric guidance, Digital bridge, Dialogue and question-answering sessions, User-friendly and accessible system.

Dedication

The authors' dedication text is awaiting confirmation. This page is retained in the supplied template's position.

Acknowledgment

We acknowledge the guidance provided during the thesis process and the public institutions whose service documents informed the knowledge corpus. We also acknowledge the developers of the open-source retrieval, embedding and local language-model software used in the implementation. Individual acknowledgements will be confirmed by the authors before submission.

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Nomenclature

BDRIS	Birth and Death Registration Information System
BRTA	Bangladesh Road Transport Authority
BM25	Best Matching 25 lexical ranking method
LLM	Large language model
MRP	Machine-readable passport
NFC	Unicode Normalization Form C
OCR	Optical character recognition
QA	Question answering
RAG	Retrieval-augmented generation
RRF	Reciprocal rank fusion

Chapter 1

Introduction

1.1 Background

Government-service information is distributed across instructions, frequently asked questions, citizen charters, forms and legal documents. A citizen’s question usually names a task rather than a document: applying for a passport, correcting a registration record or replacing a lost driving licence. A useful assistant must connect that task to the applicable procedure and preserve qualifications such as age, application stage and service type. Merely locating a document containing the same service name is insufficient.

Civic.ai, implemented in the repository as CivicRAG, addresses this problem through a curated, predominantly Bangla knowledge corpus and retrieval-augmented generation. Its final evaluated scope comprises birth and death registration, passport services and BRTA services. The records contain Bangla, English and mixed-language material, while the principal generation comparison uses Bangla questions and seeks Bangla answers. The system is a document-grounded information interface, not an integration with government transaction databases.

Retrieval-augmented generation conditions a language model on external evidence rather than relying solely on information encoded in its parameters [1]. This permits source inspection and corpus updates without retraining the answer model. However, retrieval and generation remain separate sources of error: a relevant-looking passage may concern a different procedure, and a generator may misinterpret even an appropriate passage.

1.2 Rational of the Study or Motivation

The project’s central motivation is to make collected civic-service information searchable through ordinary Bangla questions. The dataset-development effort is especially

important because operational information is not supplied as a uniform question-answer database. Scanned material, inconsistent Markdown, legal tables and historical instructions require organization before an embedding model can use them meaningfully.

The contribution is therefore not a claim to invent RAG or outperform all existing assistants. It is the development of a traceable civic-service corpus, its integration into a locally executable system and an auditable comparison with a dense-only baseline. NextRAG provides the principal architectural inspiration because it studies retrieval over Bangla and English documents in a Bangladesh-specific financial setting [2]. CivicRAG changes the domain, corpus preparation, number of retrieval channels and answer-processing logic.

Local inference was selected to support development without transmitting each question and retrieved passage to a paid hosted model. This choice also exposes a practical trade-off: compact local models are accessible, but may produce incomplete, poorly phrased or incorrectly combined instructions. The study records these limitations rather than treating local execution as a guarantee of reliability.

1.3 Problem Statement

The research problem is to develop and evaluate a source-traceable Bangla civic-service assistant from heterogeneous documents under local hardware constraints. Three technical difficulties define this problem. First, document structure and provenance must survive transformation into retrieval units. Second, the retriever must distinguish similar but non-equivalent tasks, such as passport application versus collection. Third, the generator must answer the actual question without merging conditions from unrelated passages.

The study asks:

1. How can collected civic-service files be transformed into domain-specific, auditable evidence units while preserving originals and unresolved quality flags?
2. How does the implemented hybrid retrieval and applicability-selection configuration compare with dense-only retrieval when the question set and generator settings are held fixed?
3. What do the saved outputs reveal about evidence relevance, generation behaviour and the limitations of automatic evaluation?

1.4 Objective

The implemented objectives are to organize the three service domains; preserve source-to-chunk mappings; index search-oriented text while retaining original evidence content;

implement dense and lexical retrieval with rank fusion; provide a local Bangla answer interface; and retain reproducible experiments with source IDs, routes and configuration hashes.

Objectives proposed earlier but not substantiated by the final artifacts are not reported as completed. These include training a new bilingual language model, fine-tuning BanglaBERT with Llama, conducting a citizen survey, automatically verifying credentials and evaluating general-language benchmarks such as GLUE. Model downloads and vector indexing are not described as model training.

1.5 Methodology in Brief

The offline workflow preserves original materials, reads prepared text, assigns document and chunk identifiers and creates separate search and answer representations. BGE-M3 encodes retrieval text into vectors. ChromaDB stores vectors alongside evidence text and metadata, while BM25 is constructed from the same searchable corpus.

At question time, CivicRAG searches the selected domain through both channels, combines ranks and reranks candidates. A post-P2 applicability selector screens and arranges passages before Qwen3 generates an answer. The evaluated Simple RAG baseline uses dense retrieval without hybrid fusion, reranking or that selector. Both use the same generator prompt and limits through an isolated comparison runner.

1.6 Scopes and Challenges

The final 30-question comparison contains ten passport, ten birth-registration and ten driving-licence questions. Death-registration material exists in the corpus, but the selected comparison has no death-specific questions. BRTA corpus coverage extends beyond driving licences, while those broader services are not represented in the main test set. Accordingly, results cannot establish performance across every indexed service.

The questions were reused during development. They are a development-set comparison, not an unseen benchmark. Single-turn direct questions dominate this experiment; conversational memory, long scenarios, paraphrase robustness and code-mixed input require separate evaluation. A selectable domain restricts the evidence pool, but does not by itself prove that every out-of-domain question is rejected.

1.7 Team Overview

The authors listed in the previous submission are G.Araf Ahmed, Mohammad Zunaid, Md Shihab Sarker, Fahim Foysal Abir and Tapu Mohammad Fahad. Source directories and revision history record contributions to collection and development, but do not establish a complete person-by-person division of labour. A verified responsibility table should be supplied by the authors rather than inferred from filenames.

1.8 Key Learnings and Insights

The development process demonstrates the importance of separating document relevance from answer correctness. A retrieved chunk can mention passports but describe collection rather than application. A model can faithfully repeat that chunk and still fail the user's question. Conversely, a broader answer can cover useful cases yet introduce a merged condition or exceed its output limit. The evaluation therefore distinguishes applicability, faithfulness, completeness and presentation instead of interpreting answer length or the presence of source IDs as correctness.

The Abstract is reproduced unchanged from the earlier submission. Its statements about citizen empowerment and reduced reliance on intermediaries describe the intended social outcome. This thesis does not claim a measured reduction in financial burden, office visits or intermediary use because no corresponding user-impact study was completed.

Chapter 2

Literature Review

2.1 Preliminaries

Retrieval-Augmented Generation. RAG combines an external retrieval mechanism with a generator [1]. In this implementation, retrieval selects passages at inference time and passes them to a pretrained model. No joint retriever-generator training is performed. This distinction matters because updating the corpus changes available evidence, but not the model’s learned ability to reason about that evidence.

Dense Retrieval, Lexical Retrieval and Fusion. Dense retrieval maps text to vectors and ranks candidate passages by vector proximity. BM25 instead ranks token overlap using term frequency and length normalization [3]. The channels can complement one another: semantic matching may accommodate paraphrases, while token matching can preserve important form numbers and service names. Neither channel checks whether a legal provision remains in force.

RRF combines ordered result lists rather than directly adding incompatible raw scores [4]. CivicRAG uses a weighted form. Fusion is a way to construct a hybrid result list, not the definition of hybrid retrieval itself: weighted score fusion would be another possible combination method. Its effectiveness must be tested rather than inferred from the presence of two retrievers.

Reranking and Context Selection. A cross-encoder scores a query and passage jointly, unlike independently encoded dense vectors. CivicRAG configures the multilingual BGE reranker [5]. Applicability selection is a separate, project-specific rule layer. It uses section scope and service/action vocabulary to remove or reorder passages. A reranker relevance score and a selector decision are neither probabilities of truth nor substitutes for

claim-level validation.

2.2 Review of Existing Research

NextRAG: Principal Architectural Inspiration. NextRAG addresses bilingual financial QA from Bangladeshi documents. Its method combines dense retrieval, BM25 and Dirichlet-smoothed query-likelihood retrieval, followed by weighted RRF, reranking and language-aware generation. Its reported evaluation includes reference-answer cosine similarity, F1 and inference time [2]. The shared document-language setting motivates borrowing retrieval components, but financial and civic-service applicability are not equivalent tasks.

Table 2.1: Relationship between NextRAG and the implemented CivicRAG configuration.

Aspect	NextRAG	CivicRAG
Application	Financial information	Three civic-service domains
Retrieval	Dense, BM25, Dirichlet QLM	Dense and BM25
Combination	Weighted RRF	Weighted RRF, two channels
Post-retrieval	Reranking	Reranking and heuristic applicability selection
Evidence preparation	Financial document processing	Typed evidence, source register and dual text representations
Reported comparison	Bilingual reference-answer evaluation	Paired Bangla development questions; saved outputs and diagnostic analysis
Generator identity	Includes Llama 3.1 8B	Main comparison Qwen3:8b; local llama3 is not Llama 3.1

This relationship is described as inspiration rather than replication. CivicRAG does not implement the third Dirichlet channel or reproduce the financial benchmark. The paper’s numerical results are not used as baselines for our different corpus and question set. Its reference-answer metrics are also not replaced by unlabelled similarity between two model outputs.

Multilingual Embeddings and Local Generators. BGE-M3 supports multilingual text representation and multiple retrieval modes [6]. CivicRAG uses its dense embeddings only; its sparse branch is separately implemented BM25, not BGE-M3’s learned sparse mode. This clarification prevents the embedding model’s full capabilities from being mistaken for implemented functionality.

Qwen3 supplies thinking and non-thinking modes in the same model family [7]. The selected local deployment uses the non-thinking mode for the recorded main comparison.

son. This is an inference setting, not evidence that reasoning would improve or reduce civic-answer quality. No controlled thinking-mode ablation was completed. Qwen2.5 and Llama 3 were also tested locally [8, 9]. These are specific deployed model variants, not interchangeable labels for their entire families.

Context Distraction and Diagnostic Evaluation. Research on long-context language models shows that access to relevant information does not guarantee its effective use, and that placement can affect performance [10]. The project’s selected-evidence and evidence-order trials investigate a related practical issue: whether unrelated service passages change an otherwise useful answer. They are small generation-only diagnostics. They do not demonstrate that a manually chosen context can be selected automatically for arbitrary questions.

Other Bangla RAG Approaches. TraSe investigates Bangla Wikipedia question answering using translative prompting and answer selection, with both automatic and human-in-the-loop retrieval [11]. Its distinction between automatically retrieved and human-selected contexts is useful for diagnosing our system: generation with manually selected evidence tests a different capability from end-to-end retrieval. CivicRAG’s relevant-only trials follow this diagnostic logic, but do not implement TraSe’s translation and answer-selection architecture. Their better outputs cannot be reported as automatic retrieval accuracy.

HybridRAG-BN is a recent preprint combining BM25 and BGE-M3 retrieval with generation and a fine-tuned verification stage [12]. Its use of verification highlights that retrieval relevance and answer correctness are separate problems. CivicRAG shares established retrieval components but does not train a verifier or implement that paper’s web-assisted fallback. Its competition scores are not directly comparable with our civic-service questions. These approaches inform possible future experiments, rather than retrospectively becoming components of our implementation.

Automatic RAG Evaluation. RAGAS separates aspects of retrieval and generation quality through model-assisted metrics [13]. Faithfulness checks whether response claims are supported by the supplied context. Response relevance concerns how well an answer addresses its question. A source-supported response can still answer the wrong task, so no single score is a comprehensive measure of factual correctness.

The implemented evaluator uses a local judge and documents its limitations. It does not have reviewed reference answers or gold relevance labels for the current 30 questions. Therefore reference semantic similarity, retrieval recall, MRR and nDCG are not reported

for that experiment. Reusing metric names from an earlier submission without the necessary labels would be methodologically incorrect.

2.3 Summary of Key Findings

The literature motivates multilingual retrieval, complementary lexical matching, reranking and separate assessment of context and response. Our contribution lies in the civic corpus and its traceable operationalization, with an evaluated integration of established methods. Neither the evidence selector nor the local model is presented as a newly trained state-of-the-art method.

Only sources that can be tied to the implemented method or a specific discussion are retained. Earlier draft reference lists are not inherited wholesale. Related architectures mentioned during development but not implemented are not presented as components of CivicRAG. The main comparison tests the combined configuration against dense-only retrieval; it cannot isolate the causal effect of RRF, reranking or selection individually.

Chapter 3

Requirements, Impacts and Constraints

3.1 Final Specifications and Requirements

The functional requirements are domain-scoped retrieval, Bangla-oriented generation, inspectable evidence, model selection and persistent experiment records. The system must distinguish the search representation from answer evidence, expose the passages actually supplied to generation and preserve a rollback path for changes. These requirements are implemented through domain configurations, the source register, structured response fields and Git revisions.

Table 3.1: Implemented requirements and limits of verification.

Requirement	Implementation	Verification boundary
Domain selection	Separate configurations and Chroma collections	Does not guarantee all out-of-domain rejection
Evidence traceability	Chunk IDs, metadata, source paths and available URLs	Footer is not claim-level citation verification
Bangla response	Prompt and language safety check	Fluency and semantic accuracy remain imperfect
Local execution	Ollama generation; locally cached embeddings	Latency remains substantial
Reproducible comparison	Saved questions, answers, contexts and hashes	One run per configuration; reused questions
Reversible development	Git revisions and rollback tags	Untracked artifacts need explicit preservation

The selected comparison requests Qwen3 explicitly. Some configuration files retain Llama 3.2 as a default; this does not identify the model used in saved experiments. Execution records, not an isolated default value, determine the evaluated configuration. Similarly, the selected-evidence branch allows six passages even though the legacy generation configuration retains a three-passage setting.

3.2 Societal Impact

The intended benefit is easier access to structured explanations of civic procedures. Bangla responses and inspectable evidence may reduce the effort needed to navigate multiple files. However, the reported experiments involve questions and saved model outputs rather than a deployed population study. Improvements in accessibility, trust or intermediary dependence remain intended outcomes, not measured effects.

Incorrect fees or procedure instructions can create practical harm. The interface should therefore be understood as guidance to inspect alongside official sources. A source link gives a route for checking information, but some corpus records have missing URLs or unresolved original matches. The manuscript does not imply official endorsement or authoritative legal status.

3.3 Environmental Impact

Local inference uses electricity and hardware resources. The project reuses pretrained models rather than training a foundation model, but no power measurements or carbon accounting were performed. Response time is not a substitute for energy measurement. A future study would log power draw, duration and system utilization under comparable conditions before making environmental-efficiency claims.

3.4 Ethical Issues

Three ethical concerns guide the design. First, evidence may be outdated or inapplicable even when faithfully quoted. Second, local models can combine requirements from different categories. Third, a user may interpret a confident answer as official advice. The system exposes provenance and incomplete-output warnings, but these mechanisms do not eliminate those risks.

Data curation preserves original files and records derivation rather than silently overwriting sources. Automatically detected flags indicate unresolved checks, not conclusive judgments that a file is wrong. Human review remains necessary for conflicting fees, applicability dates and OCR-sensitive legal text. Informal user preference is recorded as an observation, not converted into an invented expert accuracy score.

3.5 Standards - if applicable

The implementation uses UTF-8 text, Unicode NFC normalization, JSON/JSONL records and SHA-256 hashes. These choices support interoperable storage and change detection. They do not constitute certification under a government, security, accessibility or quality-management standard. No compliance audit is claimed.

The source and search representations retain machine-readable identifiers that permit reproducible joins across retrieval, evidence selection and evaluation. Documents without public URLs still retain local provenance; they must not be assigned fabricated URLs. Redistribution of original PDFs and personal information requires separate permission review.

3.6 Project Management Plan

The completed work proceeded through corpus preparation, a birth/death-registration implementation, extension to passport and BRTA, generation diagnostics and a matched baseline comparison. Later changes were preserved as candidates and Git revisions. The final evaluation configuration is identified through hashes rather than by describing every subsequent draft as active.

For finalization, architecture and corpus changes are separated from reporting. The outstanding automatic evaluation may finish without modifying the answer files. The thesis tables are regenerated from those artifacts, while the Abstract is checked against its immutable source. Changes that would require new answer generation are not silently included in the current comparison.

The remaining work follows dependency-based milestones rather than invented historical dates: freeze the corpus and answer revisions; finish or formally close the automatic evaluation; review paired answers; insert validated results into Chapter 5; reconcile the conclusion; and confirm administrative information before the final compile. Evaluation uses the existing local machine and Ollama models. Manuscript work uses saved artifacts and does not require repeated generation.

Completion requirement: Task PM-01: obtain team-confirmed milestone dates, responsibility assignments, available hardware and a budget/resource record. The sequence above is a plan, not a reconstructed account of hours worked. Populate this section with the confirmed schedule and budget before submission.

3.7 Risk Management

Table 3.2: Principal project risks and controls.

Risk	Existing control	Residual issue
Wrong procedure	Domain scope, reranking, applicability vocabulary	Similar headings and unclassified passages remain
Historical rules	Date fields and audit flags	Dates and legal applicability not independently verified
OCR corruption	Original-file links and review flags	Not all long legal documents manually checked
Context crowding	Whole-chunk budget and family-order fix	Broad queries may still require omitted evidence
Wrong generation	Evidence prompt and language check	No complete semantic entailment guard
Truncation	Finish-reason warning	Warning does not complete an interrupted answer
Evaluation bias	Matched inputs and disclosed development set	Reuse, self-judging and small sample remain
Version confusion	Hash manifests and rollback revisions	Index rebuilds and deployment must respect the manifest

3.8 Economic Analysis

The main generation runs use local Ollama rather than paid per-request inference. This establishes the absence of such API charges for those runs, not a zero-cost system. Hardware ownership, electricity, source preparation and review time remain costs. No invoices, time sheets or power logs support a complete monetary analysis, so no fabricated budget or savings figure is included.

At the recorded mean latency, a sequential 30-question run takes substantial wall-clock time. This limits the practicality of repeated full-batch experiments on the available machine and motivates reusing saved outputs for evaluation. Latency reduction through different hardware or smaller contexts is a future optimization, not a demonstrated economic benefit.

For a measured run, let P_{avg} be average electrical power in watts, t elapsed hours and p

the electricity price per kWh. The estimated energy charge is

$$C_{\text{energy}} = \frac{P_{\text{avg}} t}{1000} p. \quad (3.1)$$

Total project cost also includes attributable hardware, storage and documented labour. Hardware ownership cost must be distinguished from the incremental cost of one evaluation. Monetary benefits require a defensible comparator; no savings estimate is justified merely because the chatbot answers a question.

Completion requirement: Task EC-01: obtain equipment/access cost, electricity tariff, measured power and run duration, storage cost and any logged preparation/review hours. Calculate the cost table from these inputs with explicit assumptions. Citizen time savings would require an appropriate measured comparator, not generated observations.

Chapter 4

Proposed Methodology

4.1 Design Process or Methodology Overview

CivicRAG separates offline corpus/index construction from online retrieval and answer generation. The pipeline diagram in Figure 4.1 describes the selected-evidence configuration used by the main CivicRAG run. Legacy controlled-answer branches remain in the repository for rollback but are not the answer route of this comparison.

4.2 Preliminary Design or Design (Model) Specification

The P2 design emphasized birth/death registration, hybrid retrieval and controlled answer paths. Later work extended the corpus to passport and BRTA, added applicability selection and changed the main generator comparison to Qwen3. The final reported route therefore differs from the old diagrams that send every answer through a generic safe-path box.

The system is not an autonomous agent that performs government actions. Its task is single-turn retrieval and evidence-conditioned text generation. A domain is selected before retrieval. A targeted cross-domain aggregate guard handles certain multi-service questions, but comprehensive intent recognition is not claimed. Neither the lexical rules nor the selector is a separately trained classifier.

The implemented alternatives are the matched dense-only baseline and the design using hybrid retrieval, reranking and selected evidence. Both have saved 30-question executions. The relevant-only and evidence-order diagnostics isolate context handling but do not simulate a complete deployment. Their outputs verify specific design behaviours; a selector-off ablation remains unexecuted. Controlled settings appear in Section 4.4 and observed outcomes in Chapter 5.

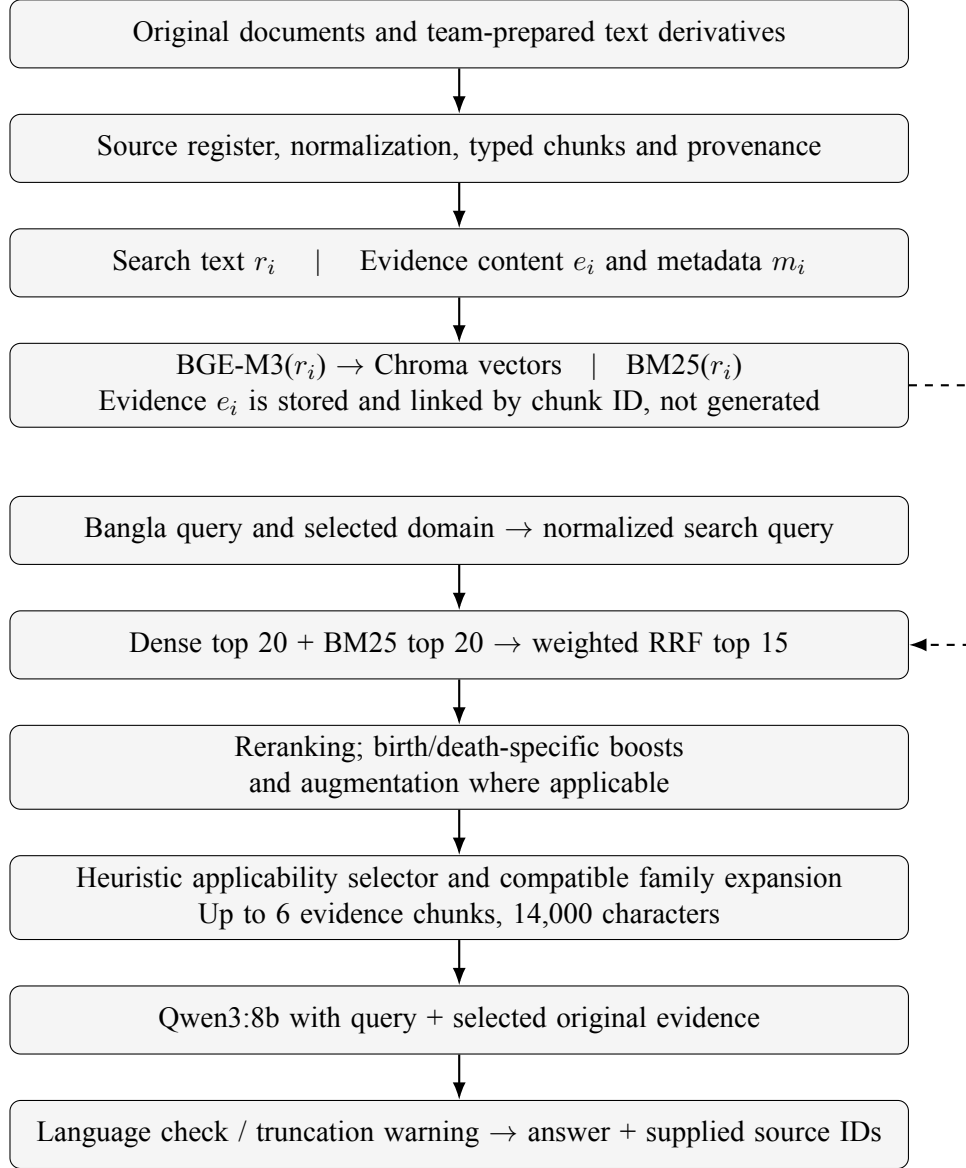


Figure 4.1: Implemented offline preparation and online selected-evidence path. The dashed link denotes access to the stored indexes, not an index-generated answer.

Completion requirement: Task DS-01: if component-level verification is required, freeze both configurations and run a selector-off comparison on the same questions, saving contexts, answers, timings and routes. Do not describe it as completed before its artifacts exist. The existing paired design comparison can be reported without inventing a simulation.

4.3 Data Collection -(If Applicable)

Source Families and Original Preservation. The birth/death collection contains structured JSON and prepared Markdown derived from FAQs, laws, rules, fees, correction processes and application guidance. Passport and BRTA were supplied as original-file directories and separately prepared Markdown. Passport derivatives are registered from `cleaned_md`; BRTA derivatives from `brta` and `fahim_brta`. Directories explicitly marked original are reference materials, not duplicate text automatically added to the answer corpus.

The register preserves document IDs, relative paths, snapshots, hashes, candidate original-file matches, inclusion history and chunk IDs. Candidate original matches do not certify textual fidelity. Some matches remain unresolved and are flagged. The distinction between collected originals and derivatives avoids counting the same information twice as independent evidence.

The corpus is a retrieval resource, not a newly annotated gold-answer benchmark. Its novelty is the organization of selected civic-service material into a source-traceable, machine-usable representation spanning the three domains. It is neither a census of all government documents nor a guarantee of current legal validity.

4.3.1 Data Cleaning

The active passport/BRTA ingestion parses simple front matter, normalizes NFC text, removes explicit authoring citation markers and selected invisible characters, and tracks headings. It preserves malformed or unrecognized material instead of making broad deletions. The work does not substantiate a universal automatic OCR repair or Bijoy-to-Unicode conversion stage for every supplied file; such claims from illustrative diagrams are not carried forward.

Separate cleanup candidates remove recognized authoring preambles and boilerplate and preserve larger structural units. Their manifests record changes and review flags. Those candidates must be distinguished from the evaluated corpus. At the audited snapshot, the active domain configurations still reference the heading-aware v2 passport/BRTA files. A

later cleanup directory alone is not evidence that its contents were activated in ChromaDB or used in the saved answers.

4.3.2 Data Transformation

Document-type-aware transformation retains question-answer pairs, headings, fee rows and process sections where the source permits. A chunk should keep its governing condition with its answer. Splitting a heading from its body can make retrieval apparently relevant while the actual content answers a different question. Heading-aware ingestion was introduced to reduce this failure, but long unstructured legal OCR remains difficult.

Each chunk has the form

$$c_i = (\text{id}_i, e_i, r_i, m_i), \quad (4.1)$$

where e_i is evidence content, r_i is search text and m_i is metadata. Search text may include aliases or section descriptors. These additions help search; they are not source facts and are not meant to become answer evidence. No query-specific reference answer is inserted into the generator merely because it resembles a test question.

4.3.3 Data Integration

Each domain has its own configured JSONL corpus and Chroma collection. Metadata preserves service context and available source URLs. A stable chunk ID joins vector results back to the source record. The runtime dense retriever obtains IDs from Chroma and reconstructs retrieval results from the in-memory chunk map, making the active JSONL and index consistency essential.

In `scripts/build_index.py`, embeddings are computed from `retrieval_text`, whereas Chroma’s documents contain `content`. The versioned ingestion script follows the same storage separation. Therefore answer evidence is stored alongside a vector; it is not itself an answer produced by Chroma. Chroma is a retrieval store, and the LLM produces the response only after evidence selection.

4.3.4 Data Reduction

Reduction is treated conservatively. FAQ duplicate handling and chunk boundaries can reduce repeated text, but semantic deduplication of the complete three-domain corpus has not been independently established. Original archives are retained. Passport and BRTA registers contain 30 and 132 derivative documents respectively; the active snapshot includes chunks from all of those registered derivatives. This statement does not imply that

every original file has a resolved one-to-one derivative match.

4.3.5 Summary of Preprocessed Data

Table 4.1: Active corpus inventory computed from configured files. Source units are prepared documents, not independent facts.

Domain	Source units	Registered originals	Chunks
Birth/death	12	Not comparable	187
Passport	30	27	361
BRTA	132	133	3161

The table is computed from the configured JSONL files, not from unrelated candidate directories. For the older birth/death corpus, source units are counted using its source-path metadata. That count is not equated to a unique original-PDF count. Across domains, chunk counts are an indexing property rather than the number of independent facts or questions.

Table 4.2: Selected unresolved active-corpus flags, counted once per derivative document. These counts may overlap.

Domain	Flag	Documents
Passport	no web address	26
Passport	original match unresolved	4
Passport	dated document review applicability	1
Passport	long document check ocr	0
BRTA	no web address	109
BRTA	original match unresolved	6
BRTA	dated document review applicability	21
BRTA	long document check ocr	8

Review flags count documents in the passport/BRTA registers or their associated chunk metadata, as indicated in the table. Multiple flags may apply to one document. A missing URL does not prove that a source is false, while a present URL does not prove that the derived text is correct. The flags identify where provenance and applicability need further review.

4.4 Implementation of Selected Design

Dense Embedding and Search. With encoder f_θ , normalized vectors are

$$v_i = \frac{f_\theta(r_i)}{\|f_\theta(r_i)\|_2}, \quad v_q = \frac{f_\theta(q)}{\|f_\theta(q)\|_2}. \quad (4.2)$$

For unit vectors, squared Euclidean distance satisfies

$$\|v_q - v_i\|_2^2 = 2 - 2v_q^\top v_i. \quad (4.3)$$

Thus distance and cosine induce the same exact ranking for normalized vectors. The index construction does not explicitly override Chroma’s distance configuration; the thesis therefore does not assert that a cosine collection was manually configured. Approximate index search can differ from an exhaustive mathematical ranking.

BGE-M3 supplies the embedding model [6]; Chroma stores the vectors and linked documents [14]. The implementation uses local cached model files and CPU embedding configuration. No corpus-specific gradient training occurs during this step.

BM25 Sparse Search. BM25 indexes tokens from r_i . Tokenization normalizes NFC, removes zero-width characters and extracts lower-case word/Bangla token sequences. The standard scoring structure is

$$\text{BM25}(q, d) = \sum_{t \in q} \text{IDF}(t) \frac{f(t, d)(k_1 + 1)}{f(t, d) + k_1(1 - b + b|d|/\bar{L})}. \quad (4.4)$$

Here $f(t, d)$ is term frequency, $|d|$ is token length and $\bar{L}$ is mean document length. The code instantiates `BM25Okapi` without explicit parameter overrides, so the installed library defaults determine k_1 , b and its IDF-floor behaviour [3, 15]. Lexical matching is a separate channel; it does not construct Chroma’s vectors.

Weighted Reciprocal Rank Fusion. The implementation retrieves at most 20 candidates from each channel, omitting BM25 results with non-positive scores. It assigns one-based ranks and computes

$$S(d) = \frac{0.55 \mathbf{1}[d \in D]}{50 + r_D(d)} + \frac{0.45 \mathbf{1}[d \in B]}{50 + r_B(d)}. \quad (4.5)$$

A missing-channel term contributes zero. As an illustrative calculation, a chunk at dense rank 2 and BM25 rank 5 receives $0.55/52 + 0.45/55 \approx 0.01876$. This is a rank-combination value, not confidence in factual accuracy. The fused list is limited to 15 before reranking. The weights are implementation settings; this thesis does not claim they are optimal for the dataset.

Reranking and Domain-Specific Logic. The configured cross-encoder scores query/search-text pairs. If loading fails, the code can add a lexical-coverage adjustment of $0.05 |T_q \cap T_d| / \max(|T_q|, 1)$ to the incoming score. The saved primary CivicRAG

candidate traces contain cross-encoder markers, establishing that the cross-encoder route was used in those records rather than merely being configured.

Birth/death-specific boosts and evidence augmentation remain part of the selected route. These are not universal learned mechanisms for passport and BRTA. This asymmetry should be disclosed when interpreting domain differences. An isolated ablation would be required to attribute an observed change to those rules.

Applicability Selection and Context Budget. The selector introduced after P2 uses service, action, product and location vocabularies. It prefers section scope over broad document titles and can reject a passport-collection passage for an application question. It also screens heading-only fragments, handles explicitly flagged historical alternatives and expands compatible fee/checklist families.

The selector is implemented in `src/generation/evidence_selection.py`; it is neither a second LLM nor a human choosing passages during normal operation. It retains at most six whole chunks within 14,000 characters. A chunk that does not fit is skipped rather than silently cut. A later ordering fix groups compatible family members near the first eligible anchor before applying the budget. Saved replay on 30 candidate sets changed the selected IDs for the passport-fee question while leaving the other 29 selections unchanged. This establishes the scope of that replay, not a general accuracy improvement.

An unknown or broad query can still retain unrelated material. Lexical applicability is not entailment, and a passage may contain mixed procedures even when its heading matches. The selector’s decisions and expanded IDs are therefore retained in a trace. Removing the selector remains a proposed ablation, not a completed experiment in this manuscript.

Answer Generation and Safety Behaviour. Qwen3:8b receives the normalized question and selected evidence through Ollama [7, 16]. The comparison uses temperature 0.05, top- p 0.9, output limit 1,200 tokens, repetition penalty 1.18, repeat window 256 and selected-route context window 8,192. Qwen3 thinking is disabled. The prompt requests evidence-grounded Bangla responses and preservation of service conditions.

The current comparison bypasses legacy controlled extraction and question-category templates. If no applicable evidence remains, it returns a clarification/insufficient-information response. A heuristic language check can replace unsuitable-language output. A length finish reason produces an explicit incomplete-output warning. These are operational safeguards, not a complete factual verifier.

The answer’s Sources footer identifies supplied context IDs, not verified supporting citations for each claim. The saved `raw_generation` field is produced after the generator’s existing response canonicalization and should not be described as untouched token-level

model output. Candidate sources and actual `answer_contexts` are stored separately to avoid evaluating unused passages as if the model had seen them.

Matched Simple RAG Baseline. The isolated runner `scripts/run_simple_matched.py` performs dense-only top-six retrieval. It does not call BM25 fusion, reranking, applicability selection, sibling expansion or controlled answers. It applies the same whole-chunk character budget, selected-route generation prompt, Qwen3 settings and language/truncation handling as the comparison configuration. The `selected_evidence=True` generator flag chooses the shared prompt; it does not invoke the selector inside the baseline.

Table 4.3: Controlled and varied elements of the main comparison.

Element	Simple RAG	CivicRAG
Corpus and questions	Same	Same
Generator and prompt	Qwen3, matched	Qwen3, matched
Dense retrieval	Top 6	Top 20
BM25 and RRF	Absent	Present
Reranker	Absent	Present
Applicability/expansion	Absent	Present
Final context limit	6 / 14,000 characters	6 / 14,000 characters
Controlled answer bypass	Absent	Absent

The comparison intentionally keeps the baseline credible. It does not weaken the generator, restrict access to the corpus or remove successful baseline answers. It measures a bundled pipeline difference and cannot independently establish the benefit of every additional component.

Chapter 5

Result Analysis

5.1 Performance Evaluation

Experimental Design. The primary comparison contains 30 paired questions with Qwen3:8b. Each question is answered once by the frozen CivicRAG selected-evidence configuration and once by the matched dense-only runner. Question records and the saved configuration/corpus hashes are checked for equality. No new answers are generated for this manuscript.

The question distribution is ten per selected domain. All 30 are labelled direct in the saved plan. Questions address documents, fees, time, application, correction, renewal, replacement and status. The balanced domain count is not a population sample of citizen demand. Because these questions were repeatedly inspected during development, the experiment estimates behaviour on a reused development set rather than unseen performance.

The outcome of a completed HTTP/model request is not equivalent to a correct answer. The route label records how the answer was produced, not whether the advice is reliable. Likewise, a finish reason of stop does not establish that every required fact was included. Completeness and correctness require inspection against applicable evidence.

Operational Metrics. For saved response times $t_1, \dots, t_n$, the reported mean is $\bar{t} = n^{-1} \sum_i t_i$. Median and nearest-rank 95th percentile summarize the distribution without assuming normality. Context count and character count describe what the model received. The Bangla-letter fraction divides Bengali-script letters by all Unicode letters in the response body after removing the source footer; it is not a measure of grammar or fluency.

Both runs completed all 30 requests. The observed mean latency is lower in the CivicRAG run, but the runs occurred separately and did not control thermal state, concurrent activity or all initialization boundaries. Therefore this observation is not reported as a proven

Table 5.1: Saved Qwen3 comparison: operational results, not correctness scores. Times are in seconds.

System	n	Mean	Median	p95	Contexts
Simple RAG	30	193.70	181.06	319.91	6.00
CivicRAG	30	181.82	161.84	285.10	4.90

speed advantage. Fewer supplied characters may contribute to response-time differences, but the experiment does not isolate that explanation.

RAGAS Evaluation Protocol and Status. The local evaluator uses RAGAS 0.2.15 [13, 17]. It evaluates actual answer contexts rather than the larger candidate list. It strips the appended source footer and otherwise preserves the response wording. The selected metrics are faithfulness, response relevance and a separately defined binary context-relevance AspectCritic.

For response statements a_j , faithfulness is conceptually the supported-statement fraction,

$$F = \frac{\sum_{j=1}^m \mathbf{1}[a_j \text{ supported by supplied context}]}{m}. \quad (5.1)$$

An empty or invalid judgment is recorded as missing, not assigned a favourable score. Response relevance reconstructs questions from the answer and compares their embeddings with the user question; the implementation uses three reconstructions and BGE-M3. It must not be relabelled as similarity to a gold answer. The custom context critic asks whether most supplied evidence is applicable to the requested service and procedure; it is not standard context precision or recall.

The local judge is Qwen3:8b, which also generated the answers. Its temperature is zero, seed 20260924, reasoning disabled, context window 16,384 and output allowance 4,096. This self-judging design is exploratory and may favour familiar output patterns. Stock English evaluation prompts applied to Bangla content add a further source of measurement uncertainty. Two simple synthetic Bangla faithfulness checks precede the batch, but passing them does not validate complex government-service judgments.

Evaluation status at manuscript snapshot: 21 of 180 metric attempts had been saved. The batch was incomplete; full-set RAGAS values are therefore not reported here. This paragraph must be refreshed from the completed artifacts before final submission. No partial mean is used as a final headline score. Failed metric calls must be disclosed with their denominators. A model can produce a highly faithful answer to an inapplicable passage; this explains why an automatic faithfulness result cannot settle the relative correctness of the two systems.

Completion requirement: Task EV-01: finalize the saved-output RAGAS batch, retain failed-call counts and refresh the results table. Task EV-02: record a consistent paired evidence-backed review before adding aggregate correctness or expert-preference scores. Both tasks feed this section and Section 5.5 without changing frozen answer files.

5.2 Analysis of Design Solutions

Selected-Evidence Diagnostics. Before the main comparison, a generation-only experiment supplied manually inspected relevant passages for three known problem questions. Qwen3 produced more useful content in these cases than when inappropriate passages dominated its context. The LLM was still invoked; the experiment did not use a hard-coded response. Retrieval and normal answer-route logic were bypassed deliberately, so the result is a diagnostic of evidence-conditioned generation rather than end-to-end retrieval performance.

An additional nine-output Qwen3 experiment used three context conditions for the same three questions: relevant-only, relevant-first with distractors, and distractor-first. Saved outputs show sensitivity to collection versus application passages, historical passport-fee information and licence cancellation versus replacement rules. The trial motivates attention to context applicability and ordering. Its small, selected sample does not establish a general causal effect across the corpus.

Observed Successes and Failure Modes. The saved baseline response to the new-passport-document question describes collection documents. The current CivicRAG response instead gives an application-oriented checklist, although its wording contains a corrupted payment phrase. For the passport-fee question, Simple RAG gives a document list rather than costs, while CivicRAG lists the four page/validity combinations present in the selected evidence. These are concrete task-alignment improvements, not a global accuracy estimate.

Conversely, the Simple RAG birth-registration timing response includes a processing period, while CivicRAG answers with the deadline for applying after birth. This is a regression in question interpretation. Both systems have problematic examples involving name correction, duplicate certificates and driving-licence status. Contradictory dates, mixed requirements and administrative instructions directed to staff can reach the answer even when the underlying source is relevant to the broad domain.

These examples support a limited conclusion: the additional pipeline can improve task alignment but does not uniformly dominate dense-only retrieval. The authors' informal domain-informed preference for broader CivicRAG responses is relevant design feedback.

Table 5.2: Qualitative paired examples from saved outputs; not adjudicated accuracy labels.

Question topic		Simple RAG observation	CivicRAG observation
Passport documents	docu- ments	Collection rather than application	Application checklist; wording issue
Passport fees		Documents instead of requested fees	Fee categories and values supplied
Birth time	processing	Includes processing period	Gives application deadline instead
Lost licence	driving li- cence	New-licence application procedure	Replacement form and GD procedure
Driving licence status	status	Unrelated address/form information	Unrelated form/marital information

It has not been converted into an aggregate expert score because no completed, consistently labelled review sheet is available.

5.3 Final Design Adjustments

The final evaluated adjustment is the compatible-family ordering fix. Prior selection could append useful sibling fee chunks after unrelated candidates, allowing the context limit to exclude them. The revised selector groups compatible family members earlier. A saved candidate replay changes one of 30 selected context lists. New generation after this fix can still vary because inference was not established as exactly deterministic across runs.

The thesis does not attribute all differences between older and newer answers to that single change. The current main comparison uses the ordering-fixed CivicRAG output, not the earlier overnight file. Proposed selector-off or dense-plus-reranker variants were discussed but are not reported as completed experiments.

5.4 Statistical Analysis

Domain-specific operational statistics expose heterogeneity hidden by a global mean. They do not measure document correctness or demonstrate performance on all services within each domain. In particular, ten birth questions do not evaluate the death-registration portion of the knowledge base.

If complete automatic scores are available, paired differences should use the same question in both systems. The evaluator computes CivicRAG minus Simple RAG on pairs with valid scores in both runs and a descriptive bootstrap interval. This interval reflects

Table 5.3: Per-domain timing and actual supplied context size for the matched comparison.

Domain	System	n	Mean seconds	Mean characters
Passport	Simple	10	176.32	5875.4
Passport	Civic	10	162.72	4220.4
Birth	Simple	10	154.34	4313.7
Birth	Civic	10	179.17	3656.7
BRTA	Simple	10	250.44	5467.6
BRTA	Civic	10	203.58	3967.0

question resampling within the reused set, not the judge’s calibration uncertainty, corpus representativeness or repeated-generation variance. A small interval should not be interpreted as eliminating those limitations.

No significance test for factual superiority is reported without a reviewed correctness variable. The earlier informal assistant estimates of correct-answer counts are excluded. Human reviewers should score relevance, correctness and completeness separately, retain their evidence basis and review system outputs without knowing which pipeline produced them where feasible.

Completion requirement: Task ST-01: after EV-01/EV-02, calculate matched per-question differences with valid-pair denominators and appropriate descriptive uncertainty. Report ties and missing judgments. An automatic score interval is not an independent correctness interval. Insert the analysis here rather than removing this required section.

5.5 Comparisons and Relationships

Table 5.4: Secondary current-configuration model runs. Language rejection counts are operational outcomes, not error rates.

Model	Requests	Mean seconds	Language rejections
Qwen3	30	181.82	0
Llama3	30	148.64	7
Qwen2.5	30	147.38	1

The secondary model runs use the same selected-evidence configuration. The artifact audit checks ordered context IDs and content equality against the Qwen3 run. Their operational outcomes provide useful engineering information, but language rejection is not equivalent to every rejected answer being factually wrong. Conversely, a non-rejected answer may still contain irrelevant or incorrect content.

Qwen3 is used as the main generator based on the project’s development observations and

its consistent use in the matched baseline. This is a pragmatic selection, not a statistically established claim that Qwen3 is the best Bangla model. Different model quantizations and inference behaviour also limit attribution of secondary differences solely to model architecture.

The P2 birth-domain retrieval table is not pooled into this experiment. It used a different question set and scope. NextRAG’s published financial QA results likewise cannot be directly ranked against these outputs. A fair external comparison would require a shared corpus, labels, model settings and protocol.

5.6 Discussions

The dataset effort creates an auditable foundation for iterative civic-service QA. A source-linked corpus makes it possible to inspect why a response used a particular procedure and to correct preparation without discarding the original files. This is a substantive contribution even when answer-generation quality remains imperfect.

The principal residual problem is applicability rather than merely vocabulary matching. A fee table, application checklist and collection FAQ can all be relevant to the word passport but serve different questions. Automatic selection reduces some distractions but remains heuristic. The generation layer then introduces another risk by combining independently plausible requirements into an incorrect statement.

The current evidence does not support 90–95% accuracy, universal superiority over Simple RAG or readiness for unsupervised official advice. It does support a developed system, a traceable multi-domain corpus, completed paired outputs and identifiable improvement/regression cases. Deployment decisions should consider both coverage and the severity of remaining errors.

Chapter 6

Conclusion

6.1 Summary of Findings

This thesis develops Civic.ai as a locally executable RAG-based civic-service information system. Its evaluated corpus covers birth/death registration, passport and BRTA through 3,709 active chunks. Search-oriented text and source evidence are kept distinct, and retrieved records remain connected to their provenance. The system integrates BGE-M3 dense retrieval, BM25, weighted RRF, reranking and a post-P2 applicability selector with local generation.

Thirty Bangla questions were answered by both the current CivicRAG configuration and a matched dense-only baseline using Qwen3. All saved requests completed, while qualitative inspection found both improvements and regressions. The broader pipeline produced more task-appropriate evidence in several application, fee and replacement cases, but did not reliably resolve all procedural distinctions. Completion, language-script statistics and response time do not establish factual accuracy.

The principal experimental limitation is the reuse of development questions and the absence of a completed independent correctness annotation. The local RAGAS evaluation is explicitly separated from factual verification and is reported only according to the available artifact status. No incomplete automatic run, unverified manual count or planned benchmark is presented as a completed result.

6.2 Contributions to the Field

The primary contribution is dataset development: organizing selected civic-service documents into domain-specific, source-traceable retrieval records while retaining originals, metadata and unresolved review flags. The second contribution is an implemented system

that makes the distinction between searchable aliases and answer evidence operational. The third is a reproducible evaluation record that separates manually selected diagnostics, automatic selected-evidence generation and a matched Simple RAG baseline.

The work adapts established retrieval techniques rather than claiming a new foundational RAG architecture. Its value is in the application-specific corpus and engineering evidence about where local Bangla service QA succeeds or fails. Applicability screening and trace preservation make failures inspectable, but are not claimed as complete solutions to hallucination or legal interpretation.

6.3 Recommendations for Future Work

The most immediate research priority is independent review of a held-out question set, including death-registration questions, broader BRTA services, paraphrases, conditional scenarios and insufficient-evidence cases. Reviewers should identify applicable passages as well as acceptable answers. This would enable retrieval recall and ranking metrics alongside answer correctness and completeness.

Data work should prioritize unresolved original matches, source dates, historical instructions and long legal OCR. Cleanup candidates should be activated only with versioned index rebuilds and renewed evaluation, rather than being assumed equivalent to the current corpus. Provenance verification and relevance labelling are different tasks and should be recorded separately.

Architecture changes should be tested through single-factor ablations, particularly selector-on versus selector-off, before introducing further retrievers or additional LLM selection stages. Better multilingual generators or independent judges may help, but their benefit must be measured under comparable conditions. Runtime optimization and concise answer composition are also needed for practical interaction.

Finally, a prospective user study could assess clarity, accessibility and practical usefulness. Such a study would be required to substantiate the social outcomes described in the preserved Abstract. Until then, Civic.ai should be presented as a developed and evaluated information system with documented scope, rather than as an official decision service or a guaranteed replacement for human guidance.

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Appendix A

How to install L^AT_EX

This appendix retains the title in the official template and describes the build requirements for this manuscript. The source uses XeTeX-compatible Unicode fonts. The verified local build uses Tectonic, which handles compilation passes and bibliography processing.

1. Install a compatible TeX toolchain. This manuscript was verified with Tectonic; XeLaTeX with BibTeX is an alternative.
2. Ensure Times New Roman and Kohinoor Bangla are available. If replacement fonts are necessary, change only the font declarations in `main.tex` and recheck all pages. The Abstract text must remain unchanged.
3. Keep `core`, `chapters`, `appendix`, `bibliography` and `generated` beside `main.tex`.
4. From the manuscript directory run `tectonic --keep-logs --keep-intermediates main.tex`.
5. Review the PDF, contents pages, equations and tables. Resolve missing characters, undefined references and overflowing text before submission.

The generated tables are included in the source package. Rebuilding the PDF does not require running the chatbot or downloading language models. Updating experimental results is a separate operation: the repository audit script reads saved artifacts and checks their configuration hashes before regenerating tables.

Appendix B

Overleaf: GitHub for L^AT_EX projects

The original template’s appendix title is retained. The manuscript is stored with its L^AT_EX source, bibliography and generated tables in the project repository. Git provides revision history and rollback points; the PDF is a compiled output rather than a substitute for editable source.

For hosted editing, transfer the source package while preserving its directories, select `main.tex` as the main document and use a Unicode-compatible compiler. Local fonts may not be available in the hosted environment. Use legally available fonts and review any layout changes. This manuscript has been compiled locally; a successful hosted build is not claimed.

Before revising the report, preserve the current Git revision. Keep manuscript edits separate from evaluation artifacts so that a reporting change does not silently alter an experiment. Compile and review the PDF before committing its source and output together. Keep incomplete scores distinct from completed results and never overwrite the unchanged Abstract while merging revisions.

Additional reproducibility notes, evaluation questions and completion requirements remain in the accompanying source package and repository. They are supporting artifacts, not replacement appendix titles or added entries in the official template’s chapter index.